

**BIHAR BOARD – 2016**

**Maths**

**Time: 3hrs 15minutes**

**Max Marks: 100**

1. The degree of a linear equation will be –

- (a) 0      (b) 2      (c) 1      (d) None of these

Ans. 1

2. Value of  $\tan \frac{\pi}{2}$  is-

- (a) 0 (b)  $\sqrt{3}$  (c)  $\frac{1}{\sqrt{3}}$       (d)  $\infty$

Ans.  $\infty$

3. Point (4, -3) lies in which quadrant?

- (a) First      (b) Second      (c) Third      (d) Fourth

Ans. Fourth

4. An angle made at the centre of the circle will be –

- (a)  $90^\circ$       (b)  $180^\circ$       (c)  $360^\circ$       (d)  $270^\circ$

Ans.  $360^\circ$

5. The mode of 3,5,2,5,7,5,8,5 will be....

- (a) 2 (b) 3      (c) 5      (d) 8

Ans. 5

6. In  $\triangle ABC$ , mid- point of AB and AC is D and E respectively, such that  $DE \parallel BC$  and  $BC = 8\text{cm}$ . Then value of DE is.....

- (a) 5cm      (b) 3cm      (c) 4cm      (d) 2cm

Ans. 4cm

7. If a coin is tossed once, the probability of getting a head will be....

- (a)  $\frac{1}{2}$       (b)  $\frac{2}{3}$       (c)  $\frac{4}{3}$       (d)  $\frac{5}{4}$

Ans.  $\frac{1}{2}$

8. Common difference of A.P 4, 10, 16, 22,...will be-

- (a) 4      (b) 6      (c) 2      (d) 8

Ans. 6

9. The corresponding sides of two similar triangles are in the ratio 4:9, then the ratio of their areas will be....

- (a)  $\frac{16}{17}$       (b)  $\frac{16}{81}$       (c)  $\frac{9}{25}$       (d)  $\frac{4}{9}$

Ans.  $\frac{16}{81}$

10.  $\frac{1}{\sqrt{3}}$  is equal to-

- (a)  $\tan 60^\circ$       (b)  $\cos 45^\circ$       (c)  $\sin 30^\circ$       (d)  $\tan 30^\circ$

Ans.  $\tan 30^\circ$

11. In quadratic equation ,  $ax^2 + bx + c = 0$  sum of roots will be-----

Ans.  $-\frac{b}{a}$

12. Value of  $\sec^2 \theta - \tan^2 \theta$  is -----

Ans. 1

13. The area of a metric curve is  $\pi(R+r)$ .....

Ans.  $(R-r)$

14. Co-ordinate points of the origin is.....

Ans. (0,0)

15. If mean of 1,4,x,5,12 is 7, then  $x = \dots\dots\dots$

Ans. 13

16. Probability of E in impossible event  $P(E) = \dots\dots\dots$

Ans. 0

17. The length of the tangents drawn on the circle from any external point are....

Ans. Equal

18. If the discriminant of a quadratic equation is zero, then both the roots will be.....

Ans. Real and equal

19. What will be the co-ordinates of mid-point of line joining the points (2,4) and (0,0).

Ans. (1, 2)

20.  $3 + 2\sqrt{5}$  is .....number.

Ans. Irrational

21. Find the zero of the polynomial  $x^2 + 7x + 10$  and test the relation between the zero and coefficient.

Ans. Given polynomial :

$$x^2 + 7x + 10 = 0$$

$$\Rightarrow x^2 + 5x + 2x + 10 = 0$$

$$\Rightarrow x(x + 5) + 2(x + 5) = 0$$

$$\Rightarrow (x + 5)(x + 2) = 0$$

Either,

$$x + 5 = 0$$

$$\therefore x = -5$$

Or,

$$x + 2 = 0$$

$$\therefore x = -2$$

So, the zeros of given polynomial are -5 and -2.

Relationship between the zeroes and coefficient:-

$$\text{Sum of zeroes} = -2 + (-5)$$

$$= -2-5$$

$$= \frac{-7}{1} = \frac{-x \text{ coefficient}}{x^2 \text{ coefficient}}$$

Product of zeroes =  $(-2)(-5)$

$$= \frac{10}{1} = \frac{\text{Constant}}{x^2 \text{ coefficient}}$$

22. Find the root of the quadratic equation  $x^2 - 3x - 10 = 0$  by factorization method.

Ans. Given equation,

$$x^2 - 3x - 10 = 0$$

$$\Rightarrow x^2 - 5x + 2x - 10 = 0$$

$$\Rightarrow x(x - 5) + 2(x - 5) = 0$$

$$\Rightarrow (x - 5)(x + 2) = 0$$

$$x - 5 = 0 ; x + 2 = 0$$

$$x = 5 \quad x = -2$$

Therefore, the required root of the given equation is 5 and -2.

23. Suppose that  $\triangle ABC$  and  $\triangle DEF$  are similar and their area are  $64\text{cm}^2$  &  $121\text{cm}^2$  respectively. If  $EF = 15.4\text{cm}$ , find  $BC$ .

Ans. Given,  $\triangle ABC \sim \triangle DEF$ ,  $EF = 15.4\text{cm}$

Area of  $\triangle ABC$  and  $\triangle DEF$  are  $64\text{cm}^2$  and  $121\text{cm}^2$  respectively.

Using similar triangle theorem, we get

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle DEF)} = \frac{BC^2}{EF^2}$$

$$\Rightarrow \frac{64}{121} = \frac{BC^2}{(15.4)^2}$$

$$\Rightarrow BC = \frac{64 \times (15.4)^2}{121}$$

$$\Rightarrow BC = \sqrt{\frac{64 \times (15.4)^2}{121}}$$

$$\therefore BC = \frac{8 \times 15.4}{11} = 11.2\text{cm}$$

24. Prove that the tangents drawn at the ends of a diameter of a circle are parallel.

Ans. Let AB be the diameter of a circle with centre O. The tangents PQ and RS are drawn at point A and B respectively.

We know that a tangent at any point of a circle is perpendicular to the radius through the point of contact.

$\therefore OA \perp PQ$  and  $OB \perp RS$

So,

$$\angle OBR = 90^\circ$$

$$\angle OBS = 90^\circ$$

$$\angle OAP = 90^\circ$$

$$\angle OAQ = 90^\circ$$

Therefore, we get

$$\angle OBR = \angle OAQ$$

$$\angle OBS = \angle OAP$$

Also, these are the pair of alternate interior angles.

Now,

Since the alternate interior angles are equal, we can say that the lines PQ and RS are parallel to each other.

Hence proved.

25. Find the distance between the points (-5, 7) and (-1, 3).

Ans. Let the given points be A (-1, 3) and B (-5, 7).

Then, AB=

$$\sqrt{(-5 - (-1))^2 + (7 - 3)^2}$$

$$= \sqrt{(-4)^2 + (4)^2}$$

$$= \sqrt{16 + 16}$$

$$= \sqrt{32}$$

$$= 4\sqrt{2}$$

Hence, the distance between the given points is a.

26. Find the area of a sector of a circle with radius 6 cm if angle of the sector is  $60^\circ$ .

Ans. Given, radius (r) = 6cm

Angle of sector ( $\theta$ ) =  $60^\circ$

We know,

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

$$\frac{60^\circ}{360^\circ} \times \frac{22}{7} \times (6)^2$$

$$\begin{aligned} \therefore \text{Area of sector} &= \frac{1}{6} \times \frac{22}{7} \times 36 \\ &= 18.86 \text{ cm}^2 \end{aligned}$$

27. Edges of three cubes are 3cm ,4cm and 5cm respectively. The three have been melted and a new cube is formed, find the edge of the new cube.

Ans. Given, cubes with edge 3cm ,4cm and 5cm.

So,

Volume of each cube will be

$$\begin{aligned} &= [(5 \times 5 \times 5), (4 \times 4 \times 4), (3 \times 3 \times 3)] \\ &= (125, 64, 27) \text{ cm}^3 \end{aligned}$$

$$\therefore \text{Total volume will be } (125+64+27) = 216 \text{ cm}^3$$

Thus, this will be the volume new formed cube .

Now,

$$\text{Edge of cube} = \sqrt[3]{\text{Volume}} = \sqrt[3]{216} = 6 \text{ cm}$$

Hence, the edge of new cube formed is 6cm.

28. If  $\sec \theta = \frac{13}{12}$ , find  $\cot \theta$ .

Ans. Given,

$$\sec \theta = \frac{13}{12} = \frac{h}{b}$$

Using Pythagoras theorem,

$$\begin{aligned} p^2 &= h^2 - b^2 \\ \Rightarrow p^2 &= (13)^2 - (12)^2 \\ \Rightarrow p^2 &= 169 - 144 \\ \Rightarrow p^2 &= 25 \\ \therefore p &= 5 \end{aligned}$$

$$\therefore \cot \theta = \frac{b}{p} = \frac{12}{5}$$

29. Show that  $\tan 45^\circ \cdot \tan 23^\circ \cdot \tan 67^\circ = 1$

Ans. Given ,

$$\begin{aligned} \tan 45^\circ \cdot \tan 23^\circ \cdot \tan 67^\circ &= 1 \\ \Rightarrow 1 \cdot \tan(90^\circ - 67^\circ) \cdot \tan 67^\circ &= 1 \\ \Rightarrow \cot 67^\circ \cdot \tan 67^\circ &= 1 \\ \Rightarrow \frac{1}{\tan 67^\circ} \cdot \tan 67^\circ &= 1 \\ \therefore 1 &= 1 \end{aligned}$$

Hence proved.

30. Find the area of the triangle whose vertices are (-5,-1) (3,-5) (5,2).

Ans. Given points  $(-5,-1) = (x_1, y_1)$ ,

$$(3,-5) = (x_2, y_2) \text{ and}$$

$$(5, 2) = (x_3, y_3)$$

We know,

$$\text{Area of triangle} = \frac{1}{2} [x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$$

$$\begin{aligned}
 &= \frac{1}{2}[-5(-5-2) + 3(2-(-1)) + 5(-1-(-5))] \\
 &= \frac{1}{2}[-5(-5-2) + 3(2+1) + 5(-1+5)] \\
 &= \frac{1}{2}(-5 \times (-7) + 3 \times 3 + 5 \times 4) \\
 &= \frac{1}{2}(35 + 9 + 20) \\
 &= \frac{1}{2} \times 64 \\
 &= 32 \text{ sq. units}
 \end{aligned}$$

31. Which term of the A.P : 3, 8, 13, 18, ....is 78?

Ans. Given A.P: 3,8,13,18,.....,78

First term( $a$ )=3,

Common difference ( $d$ )=  $a_2 - a_1 = 8-3 = 5$

Let 78 be the  $n$ th term,

Then, using the formula,

$$a_n = a + (n-1)d$$

$$\Rightarrow 78 = 3 + (n-1)5$$

$$\Rightarrow 75 = (n-1)5$$

$$\Rightarrow 75 = 5n - 5$$

$$\Rightarrow 80 = 5n$$

$$\Rightarrow n = \frac{80}{5}$$

$$\therefore n = 16$$

Hence, 78 is 16<sup>th</sup> term of the given A.P.

32. Find the H.C.F and L.C.M of 8, 9 and 25 by prime factorization method.

Ans. Here, given numbers are 8, 9 and 25.

Using factorization method,

$$8 = 1 \times 2 \times 2 \times 2$$

$$9 = 1 \times 3 \times 3$$

$$25 = 1 \times 5 \times 5$$

$$\therefore \text{HCF} = 1 \text{ and}$$

$$\text{LCM} = 1 \times 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 5 = 1800$$

33. Find the root of the quadratic equation  $x^2 - 2\sqrt{2}x + 1 = 0$ .

Ans. Given equation :

$$x^2 - 2\sqrt{2}x + 1 = 0$$

Comparing it with standard form  $ax^2 + bx + c$ ,

We get,  $a = 1$ ,  $b = -2\sqrt{2}$  and  $c = 1$

Now, putting these values in the formula,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow x = \frac{-(-2\sqrt{2}) \pm \sqrt{(-2\sqrt{2})^2 - 4 \times 1 \times 1}}{2 \times 1}$$

$$\Rightarrow x = \frac{2\sqrt{2} \pm \sqrt{(4 \times 2) - 4}}{2}$$

$$\Rightarrow x = \frac{2\sqrt{2} \pm \sqrt{4}}{2}$$

$$\Rightarrow x = \frac{2\sqrt{2} \pm 2}{2}$$

$$\Rightarrow x = \sqrt{2} \pm 1$$

$$\therefore x = \sqrt{2} + 1 \text{ or } \sqrt{2} - 1$$

34. Solve  $2x + 3y = 11$  and  $2x - 4y = -24$  and find the value of  $m$  if  $y = mx + 3$ .

Ans. Given equations are

$$2x + 3y = 11 \text{ -----(1)}$$

$$2x - 4y = -24 \text{ -----(2)}$$

From equation (2), we get

$$2x - 4y = -24$$

$$2(x - 2y) = 2(-12)$$

$$x - 2y = -12$$

$$x = -12 + 2y$$

Now,

Put  $x = -12 + 2y$  in equation (1)

$$2x + 3y = 11$$

$$2(-12 + 2y) + 3y = 11$$

$$-24 + 4y + 3y = 11$$

$$7y = 11 + 24$$

$$7y = 35$$

$$y = \frac{35}{7}$$

$$\therefore y = 5$$

So,

$$x = -12 + 2y$$

$$x = -12 + 2(5)$$

$$x = -12 + 10$$

$$\therefore x = -2$$

Now,

Putting the value of  $x$  and  $y$  in the given equation,

$$y=mx+3$$

$$5=m(-2)+3$$

$$5=-2m+3$$

$$-2m=5-3$$

$$-2m=2$$

$$m=\frac{2}{-2}$$

$$\therefore m=-1$$

35. In A.P,  $a = 5$ ,  $d = 3$  and  $a_n = 50$ . Find  $n$  and  $S_n$ .

Ans. Given,

$$a = 5$$

$$d = 3$$

We know,

$$a_n = a + (n-1)d$$

$$\Rightarrow 50 = 5 + (n-1)3$$

$$\Rightarrow 50 = 5 + 3n - 3$$

$$\Rightarrow 3n = 50 - 2$$

$$\Rightarrow 3n = 48$$

$$\therefore n = \frac{48}{3} = 16$$

And,

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$\therefore S_{16} = \frac{16}{2}[2 \times 5 + (16-1)3]$$

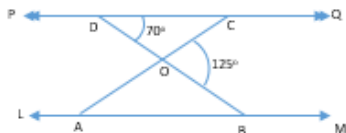
$$= 8(10+15 \times 3)$$

$$= 8(10+45)$$

$$= 8 \times 55$$

$$= 440$$

36. In the given figure, if  $LM \parallel PQ$ ,  $\angle BOC = 125^\circ$  and  $\angle CDO = 70^\circ$ , then find  $\angle DCO$  and  $\angle DOC$ .



Ans. Given,  $LM \parallel PQ$ ,  $\angle BOC = 125^\circ$  and  $\angle CDO = 70^\circ$

Here,

$$\angle BOC + \angle DOC = 180^\circ \quad [\text{Being angles of a straight line}]$$

$$\Rightarrow 125 + \angle DOC = 180^\circ$$

$$\therefore \angle DOC = 55^\circ$$

Now, in  $\triangle DOC$

$$\angle DOC + \angle DCO + \angle CDO = 180 \quad [\because \text{Sum of angles of a triangle} = 180^\circ]$$

$$\Rightarrow 55 + \angle DCO + 70 = 180$$

$$\Rightarrow \angle DCO = 180 - 125$$

$$\therefore \angle DCO = 55^\circ$$

37. The co-ordinates of the vertices of a triangle are (5,0), (3, 2) and (4,7). Find the co-ordinates of the centroid of the triangle.

Ans. Given points are: (5,0) =  $(x_1, y_1)$ ,

$$(3,2) = (x_2, y_2) \text{ and}$$

$$(4, 7) = (x_3, y_3)$$

$$\therefore \text{Centroid of the triangle} = \left\{ \frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right\}$$

$$= \left( \frac{5+3+4}{3}, \frac{0+2+7}{3} \right)$$

$$= \left( \frac{12}{3}, \frac{9}{3} \right)$$

$$= (4, 3)$$

38. A bag contains 3 red balls and 5 black balls. A ball is drawn at random from the bag.

What is the probability that the ball drawn is red?

Ans. Given,

Number of red balls = 3

Number of black balls = 5

Total number of balls =  $5+3 = 8$

$\therefore$  Probability of drawing red balls =  $\frac{\text{Number of red balls}}{\text{Total number of balls}} = \frac{3}{8}$

39. Prove that:  $\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} = 2 \sec A$

Ans. L.H.S =

$$\begin{aligned} & \frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} \\ &= \frac{\cos^2 A + (1 + \sin A)^2}{\cos A(1 + \sin A)} \\ &= \frac{\cos^2 A + 1 + 2 \sin A + \sin^2 A}{\cos A(1 + \sin A)} \\ &= \frac{1 + 1 + 2 \sin A}{\cos A(1 + \sin A)} \quad [\because \sin^2 A + \cos^2 A = 1] \\ &= \frac{2(1 + \sin A)}{\cos A(1 + \sin A)} \\ &= \frac{2}{\cos A} \\ &= 2 \sec A \\ &= R.H.S \end{aligned}$$

40. Prove that:  $\frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} = \tan \theta$

Ans. L.H.S=

$$\begin{aligned} & \frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} \\ &= \frac{\sin \theta (1 - 2 \sin^2 \theta)}{\cos \theta (2 \cos^2 \theta - 1)} \\ &= \frac{\sin \theta (1 - \sin^2 \theta - \sin^2 \theta)}{\cos \theta (\cos^2 \theta + \cos^2 \theta - 1)} \quad [\because \sin^2 \theta + \cos^2 \theta = 1] \\ &= \frac{\sin \theta (\cos^2 \theta - \sin^2 \theta)}{\cos \theta (\cos^2 \theta - \sin^2 \theta)} \\ &= \frac{\sin \theta}{\cos \theta} \\ &= \tan \theta \\ &= R.H.S. \text{ Proved} \end{aligned}$$

41. Find mode of the given data:

| Class-interval  | 0-20 | 20-40 | 40-60 | 60-80 | 80-100 | 100-120 |
|-----------------|------|-------|-------|-------|--------|---------|
| No. of students | 10   | 35    | 52    | 61    | 38     | 29      |

Ans. Since 61 is the highest frequency, so the mode class interval is 60-80.

Now,

$$l = 60$$

$$f_1 = 61$$

$$f_0 = 52$$

$$f_2 = 38$$

$$h = 20$$

Using formula for mode,

Mode =

$$\begin{aligned}
 & l + \left( \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h \\
 &= 60 + \left( \frac{61 - 52}{2 \times 61 - 52 - 38} \right) \times 20 \\
 &= 60 + \left( \frac{9}{122 - 90} \right) \times 20 \\
 &= 60 + \frac{9}{32} \times 20 \\
 &= 60 + 5.625 \\
 &= 65.625 (\text{approximately})
 \end{aligned}$$

42. Find mean of the given data:

|                |      |       |       |       |       |
|----------------|------|-------|-------|-------|-------|
| Class-interval | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 |
| No. of person  | 5    | 10    | 25    | 5     | 11    |

Ans.

| Class-interval | Mid-point(x) | Frequency(f)  | fx               |
|----------------|--------------|---------------|------------------|
| 0-10           | 5            | 5             | 25               |
| 10-20          | 15           | 10            | 150              |
| 20-30          | 25           | 25            | 625              |
| 30-40          | 35           | 5             | 175              |
| 40-50          | 45           | 11            | 495              |
|                |              | $\sum f = 56$ | $\sum fx = 1470$ |

$$\therefore \text{Mean} = \frac{\sum fx}{\sum f} = \frac{1470}{56} = 26.25$$

43. A wire is in the form of a circle whose radius is 84cm. Find the area of the square that can be formed by bending the wire.

Ans. Given, radius of circle (r) = 84cm

$$\therefore \text{Circumference of circle} = 2\pi r$$

$$= 2 \times \frac{22}{7} \times 84$$

$$= 528\text{cm}$$

Now,

Circumference of circle = Perimeter of square

$$\Rightarrow 528 = 4 \times \text{side}$$

$$\Rightarrow \text{side} = \frac{528}{4}$$

$$\therefore \text{side} = 132\text{cm}$$

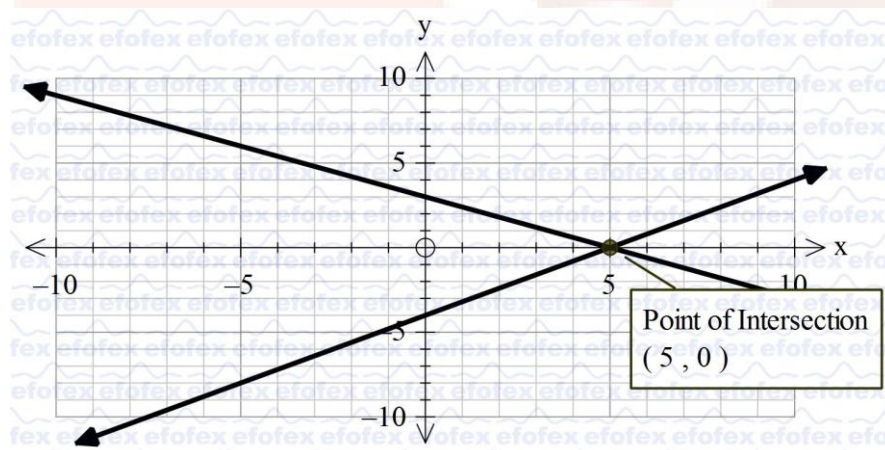
$$\therefore \text{Area of square} = (\text{side})^2$$

$$= (132)^2 = 17424\text{cm}^2$$

44. Solve graphically:

$$4x - 5y = 20 \text{ and } 3x + 5y = 15$$

Solution:



45. A vertical tower subtends a right angle on the top of 5m high flagstaff. If distance between them is 12m, find the height of the tower.

Ans. Let AB be the flag staff and CD be the vertical tower.

$$AB = 5\text{m} = EC$$

$$BC = 12\text{m} = AE$$

$$\text{let } \angle EAC = x$$

then  $\angle DAE = (90 - x)$

In  $\triangle ABC$  we have,

$$\tan x = \frac{AB}{BC}$$

$$\Rightarrow \tan x = \frac{5}{12}$$

$$\Rightarrow \frac{5}{\tan x} = 12 \dots\dots (i)$$

In  $\triangle ADE$  we have,

$$\tan(90 - x) = \frac{DE}{AE}$$

$$\Rightarrow \cot x = \frac{DE}{12}$$

$$\Rightarrow \frac{DE}{\cot x} = 12 \dots\dots (ii)$$

Now,

From (i) and (ii) we get,

$$\frac{5}{\tan x} = \frac{DE}{\cot x}$$

$$\Rightarrow \frac{5}{\frac{5}{12}} = \frac{DE}{\frac{12}{5}}$$

$$\Rightarrow 5 \times \frac{12}{5} = DE \times \frac{5}{12}$$

$$\Rightarrow 12 = DE \times \frac{5}{12}$$

$$\therefore DE = \frac{144}{5} = 28.8m$$

Therefore, height of the tower =  $DE + EC = 28.8 + 5 = 33.8m$

OR

A 10m high flag stands on a tower. From a point on the ground, the angle of elevation at the foot and the top of the flag are  $30^\circ$  and  $60^\circ$  respectively. Find the height of the tower.

Ans. Let AD be the flag and BD be the tower.

Given , AD = 10cm, let BD be h cm.

In  $\triangle BCD$ ,

$$\tan 30^\circ = \frac{BD}{BC}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{BC}$$

$$\Rightarrow BC = \sqrt{3}h \dots (i)$$

Again , in  $\triangle ABC$

$$\tan 60^\circ = \frac{AB}{BC}$$

$$\Rightarrow \sqrt{3} = \frac{10 + h}{BC}$$

$$\Rightarrow BC = \frac{10 + h}{\sqrt{3}} \dots (ii)$$

From equation (i) and (ii), we get

$$\sqrt{3}h = \frac{(10 + h)}{\sqrt{3}}$$

$$\Rightarrow 3h = (10 + h)$$

$$\Rightarrow 2h = 10$$

$$\therefore h = 5\text{cm}$$

46. In  $\triangle ABC$ , side BC stands on side AD in such a way that  $\angle ADC = \angle BAC$ . then show that  $AC^2 = BC \times CD$

Ans. Here, in  $\triangle ADC$  and  $\triangle BAC$ ,

$\angle ACD = \angle BCA$  (Common angle)

$\therefore \triangle ADC \sim \triangle BAC$  (By AA similarity criterion)

We know that corresponding sides of similar triangles are in proportion.

$$\frac{AC}{BC} = \frac{CD}{AC}$$

$$\therefore AC^2 = BC \times CD.$$

47. Draw a tangent to a circle of radius 4cm from a point on the concentric circle of radius 6cm and measure its length.

Ans. Steps of construction:

1. Draw two concentric circles  $C_1$  and  $C_2$  with common Centre O and radius 4 cm and 6 cm.
2. Take any point P on outer circle  $C_2$  and join OP.
3. Draw the bisector of OP which bisects OP at M'.
4. Taking M' as centre and OM' as radius draw a dotted circle which cuts the inner circle  $C_1$  at two points Q & R.
5. Join PQ & PR.

Thus, PQ & PR are the required tangents.

On measuring PQ & PR,

$$PQ = PR = 4.4 \text{ cm}$$

OR

Construct a right angle triangle whose hypotenuse is 8cm and the difference between the other two sides is 2cm. Also construct a circle around the triangle.

Ans. Steps of construction:

1. Draw a horizontal line passing from point A.
2. Construct an angle of  $45^\circ$  at point A and pull it back.
3. Draw an intercept of the difference (given)
4. Taking Q as vertex of triangle, draw a hypotenuse in respect of horizontal line B cut at R.
5. Draw an angle of  $45^\circ$  at R on the horizontal line that meets the line QA at P.

Hence, the required triangle is  $\triangle PQR$